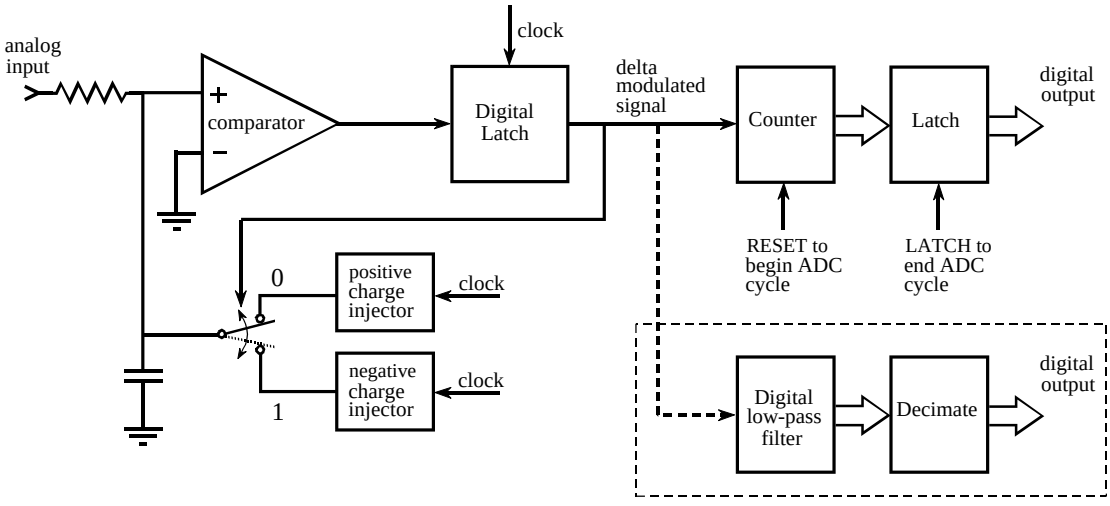

FIGURE 3-19
Block diagram of a delta-sigma analog-to-digital converter. In the simplest case, the pulses from a delta modulator are counted for a predetermined number of clock cycles. The output of the counter is then latched to complete the conversion. In a more sophisticated circuit, the pulses are passed through a digital low-pass filter and then resampled (decimated) to a lower sampling rate.

and low voltages corresponding to the digital ones and zeros average out to form the correct analog voltage. For example, suppose that the ones and zeros are represented by 5 volts and 0 volts, respectively. If 80% of the bits in the data stream are *ones*, and 20% are *zeros*, the output of the low-pass filter will be 4 volts.

This method of transforming the single bit data stream back into the original waveform is important for several reasons. First, it describes a slick way to replace the counter in the delta-sigma ADC circuit. Instead of simply counting the pulses from the delta modulator, the binary signal is passed through a *digital* low-pass filter, and then *decimated* to reduce the sampling rate. For example, this procedure might start by changing each of the ones and zeros in the digital stream into a 12 bit sample; ones become a value of 4095, while zeros become a value of 0. Using a digital low-pass filter on this signal produces a digitized version of the original waveform, just as an analog low-pass filter would form an analog recreation. Decimation then reduces the sampling rate by discarding most of the samples. This results in a digital signal that is equivalent to direct sampling of the original waveform.

This approach is used in many commercial ADC's for digitizing voice and other audio signals. An example is the National Semiconductor ADC16071, which provides 16 bit analog-to-digital conversion at sampling rates up to 192 kHz. At a sampling rate of 100 kHz, the delta modulator operates with a clock frequency of 6.4 MHz. The low-pass digital filter is a 246 point FIR, such as described in Chapter 16. This removes all frequencies in the digital data above 50 kHz, ½ of the eventual sampling rate. Conceptually, this can be